1 **(a)** $y = kx^2$ or $36 = k(3^2)$ **[M1]**

$4x^2$ **[A1]**

(b) 100 **[B1]**

[Total Marks: 3]

2 **(a)** Sets up equation $y = kx$ and substitutes $x = 5, y = 15$ to get $15 = 5k$ **[M1]**

$y = 3x$ **[A1]**

(b) Substitutes $x = 12$ into their equation from part (a), e.g. $y = 3 \times 12$ **[M1]**

36 **[A1]**

[Total Marks: 4]

3 **(a)** Substitutes into the proportion equation to find the constant, e.g. $135 = k(1+2)^3$ **[M1]**

$5(x+2)^3$ **[A1]**

(b) 40 **[B1]**

(c) Sets up the equation $320 = 5(x+2)^3$ and simplifies to $(x+2)^3 = 64$ **[M1]**

2 **[A1]**

[Total Marks: 5]

4 **(a)** Sets up $P = k_1\sqrt{Q}$ and substitutes
$Q = 9, P = 12 \implies 12 = k_1\sqrt{9} \implies 12 = 3k_1 \implies k_1 = 4$, so $P = 4\sqrt{Q}$ **[M1]**

Sets up $Q = \dfrac{k_2}{R^3}$ and substitutes $R = 2, Q = 2 \implies 2 = \dfrac{k_2}{2^3} \implies 2 = \dfrac{k_2}{8} \implies k_2 = 16$, so $Q = \dfrac{16}{R^3}$ **[M1]**

Substitutes Q into the P equation: $P = 4\sqrt{\dfrac{16}{R^3}}$ **[M1]**

Simplifies the square root: $P = 4 \times \dfrac{4}{\sqrt{R^3}}$ **[M1]**

$P = \dfrac{16}{R^{1.5}}$ or $P = 16R^{-1.5}$ or $P = \dfrac{16}{R\sqrt{R}}$ **[A1]**

(b) Substitutes $R = 4$ into their formula: $P = \dfrac{16}{4\sqrt{4}}$ or $P = \dfrac{16}{8}$ **[M1]**

2 **[A1]**

[Total Marks: 7]

5 **(a)** Sets up equation $y = \dfrac{k}{x^2}$ **[M1]**

Substitutes $x = 4, y = 3$ to get $3 = \dfrac{k}{4^2} \implies k = 3 \times 16 = 48$ **[M1]**

$y = \frac{48}{x^2}$ **[A1]**

(b) Sets $0.75 = \frac{48}{x^2}$ **[M1]**

Rearranges to $x^2 = \frac{48}{0.75} = 64$ **[M1]**

$x = 8, x = -8$ (both required for final mark) **[A1]**

[Total Marks: 6]

6 (a) $M = \frac{k}{\sqrt{N}}$ or $15 = \frac{k}{\sqrt{16}}$ **[M1]**

$\frac{60}{\sqrt{N}}$ **[A1]**

(b) $\frac{60}{\sqrt{100}}$ or $\frac{60}{10}$ **[M1]**

6 **[A1]**

(c) $1.2 = \frac{60}{\sqrt{N}}$ **[M1]**

$\sqrt{N} = \frac{60}{1.2}$ or $\sqrt{N} = 50$ **[M1]**

2500 **[A1]**

[Total Marks: 7]

7 (a) Substitutes into the proportion equation to find the constant, e.g. $4 = \frac{k}{\sqrt{36}}$ **[M1]**

$\frac{24}{\sqrt{Q}}$ **[A1]**

(b) 2 **[B1]**

(c) Sets up the equation $1.5 = \frac{24}{\sqrt{Q}}$ leading to $\sqrt{Q} = 16$ **[M1]**

256 **[A1]**

(d) Identifies the new value of Q is $4Q$, or uses numerical values (e.g. Q increases from 36 to 144) **[M1]**

Evaluates the new value of P as $\frac{1}{2}P$, or calculates the new P is 2 when original was 4 **[M1]**

50 **[A1]**

[Total Marks: 8]

8 (a) Tests products or ratios, e.g., shows $x^2y = 300$ for all pairs or identifies $k = 300$ **[M1]**

Identifies $y \propto \frac{1}{x^2}$ as the correct model **[A1]**

$\frac{300}{x^2}$ **[A1]**

(b) Uses 1.5 multiplier and finds $(1.5)^3$ or 3.375 **[M1]**

Identifies increase factor as $3.375 - 1$ or 2.375 **[M1]**

237.5 **[A1]**

[Total Marks: 6]

9 (a) (i) Recognises that x^2y is constant (128), or identifies the model as $y \propto \frac{1}{x^2}$ **[M1]**

Sets up the equation $y = \frac{k}{x^2}$ and correctly calculates $k = 128$ **[M1]**

$\frac{128}{x^2}$ **[A1]**

(ii) 1.28 **[B1]**

(b) Sets up the correct scale factor equation for B, e.g. $m^3 = 1.728$ or $\left(1 + \frac{x}{100}\right)^3 = 1.728$ **[M1]**

Calculates the multiplier $m = \sqrt[3]{1.728} = 1.2$ **[M1]**

20 **[A1]**

[Total Marks: 7]